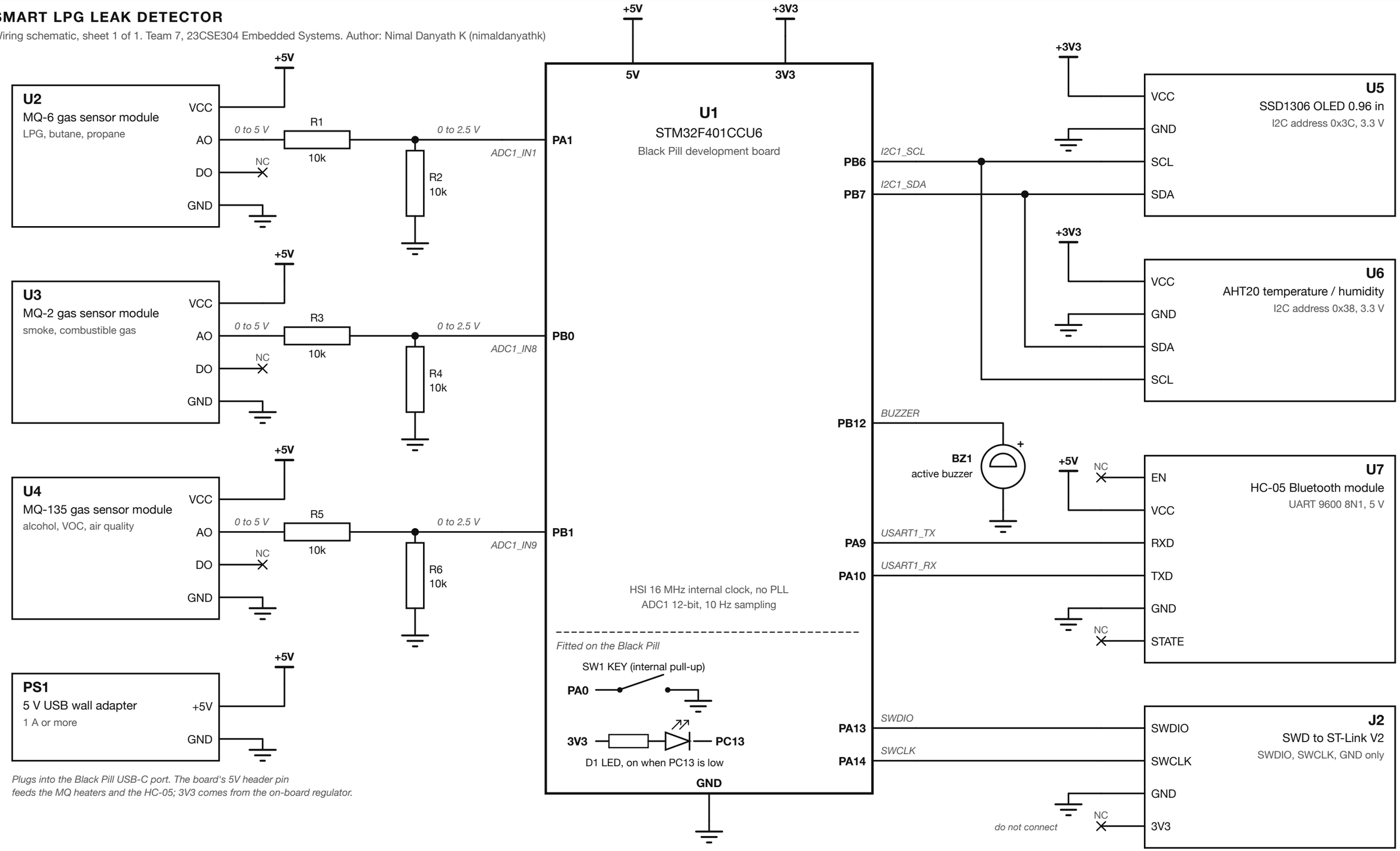
SMART LPG LEAK DETECTOR

Wiring schematic, sheet 1 of 1. Team 7, 23CSE304 Embedded Systems. Author: Nimal Danyath K (nimaldanyathk)



NOTES

1. R1 to R6: 10 kOhm. Each pair halves the 0 to 5 V sensor output, so the ADC pins see at most 2.5 V (3.3 V limit).
2. MQ-6, MQ-2, MQ-135 and HC-05 run from the 5 V rail. Each MQ heater draws about 150 mA, so use a 5 V adapter rated 1 A or more.
3. SSD1306 and AHT20 run from the Black Pill 3.3 V regulator and share I2C1. Internal pull-ups on PB6/PB7 are enabled in firmware; none are fitted externally.
4. SWD: connect SWDIO, SWCLK and GND only. Leave the ST-Link 3.3 V pin open while the board is powered over USB.
5. SW1 (KEY) and D1 (LED) are fitted on the Black Pill. D1 is active low. DO pins of the MQ modules are not used.
6. Firmware: HSI 16 MHz, sensors sampled at 10 Hz, USART1 9600 8N1. Alarm logic is described in the README.

● wire junction. Crossing wires without a dot are not connected. NC = not connected.

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SMART LPG LEAK DETECTOR		SHEET **1 / 1**	SIZE **A3**
TITLE: Wiring schematic, STM32F401CCU6 Black Pill		REV: **A**	
DRAWN BY: **Team 7**	AUTHOR: Nimal Danyath K	DATE: 2026-10-15	
PROJECT: 23CSE304 Embedded Systems, Amrita Vishwa Vidyapeetham		DOCUMENT NO: LPG-SCH-001	